

Overview of Variational autoencoder

Subject: Variational autoencoder

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \| p(z))$$

1.

Unfortunately, the computation of $p_\theta(z|x)$ is expensive and in most cases intractable. To speed up the calculus to make it feasible, it is necessary to introduce a further function to approximate the posterior distribution as

2.

In the vanilla variational autoencoder, z is usually taken to be a finite-dimensional vector of real numbers, and $p_\theta(x|z)$ to be a Gaussian distribution. Then $p_\theta(x)$ is a mixture of Gaussian distributions. It is now possible to define the set of the relationships between the input data and its latent representation as

3.

$$\mathcal{L}_{\theta, \phi} = \mathbb{E}_{x \sim \text{Real}} [\|x - D_\theta(E_\phi(x))\|_2^2] + d(\mu(dz), E_\phi(\text{Real}))^2 \quad \mathbb{E}_{x \sim \text{Real}} [\|x - D_\theta(E_\phi(x))\|_2^2] + d(\mu(dz), E_\phi(\text{Real}))^2$$

4.

Now, define the evidence lower bound (ELBO): $\mathcal{L}_{\theta, \phi}(x) := \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x, z) q_\phi(z|x)] = \ln p_\theta(x) - D_{KL}(q_\phi(\cdot|x) \| p_\theta(\cdot|x))$

Maximizing the ELBO $\theta^*, \phi^* = \arg\max_{\theta, \phi} \mathcal{L}_{\theta, \phi}(x)$ is equivalent to simultaneously maximizing $\ln p_\theta(x)$ and minimizing $D_{KL}(q_\phi(z|x) \| p_\theta(z|x))$. That is, maximizing the log-likelihood of the observed data, and minimizing the divergence from the approximate posterior $q_\phi(\cdot|x)$ to the exact posterior $p_\theta(\cdot|x)$. The form given is not very convenient for maximization, but the following, equivalent form, is: $\mathcal{L}_{\theta, \phi}(x) = \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln p_\theta(x|z)] - D_{KL}(q_\phi(\cdot|x) \| p_\theta(\cdot))$ where $\ln p_\theta(x|z)$ is implemented as $-\frac{1}{2} \|x - D_\theta(z)\|_2^2$, since that is, up to an additive constant, what $x|z \sim N(D_\theta(z), I)$ yields. That is, we model the distribution of x conditional on z to be a Gaussian distribution

centered on $D_{\theta}(z)$. The distribution of $q_{\phi}(z|x)$ and $p_{\theta}(z)$ are often also chosen to be Gaussians as $z|x \sim N(E_{\phi}(x), \sigma_{\phi}(x)^2 I)$ and $z \sim N(0, I)$, with which we obtain by the formula for KL divergence of Gaussians: $L_{\theta, \phi}(x) = -\frac{1}{2} E_{z \sim q_{\phi}(\cdot|x)} [\log \frac{p_{\theta}(z)}{q_{\phi}(z)}] = -\frac{1}{2} (N \sigma_{\phi}(x)^2 + \|E_{\phi}(x)\|^2 - 2N \ln \sigma_{\phi}(x)) + \text{Const}$. Here N is the dimension of z . For a more detailed derivation and more interpretations of ELBO and its maximization, see its main page.

5.

the sliced Wasserstein distance used by S Kolouri, et al. in their VAE the energy distance implemented in the Radon Sobolev Variational Auto-Encoder the Maximum Mean Discrepancy distance used in the MMD-VAE the Wasserstein distance used in the WAEs kernel-based distances used in the Kernelized Variational Autoencoder (K-VAE)

6.

the usual reconstruction error part which seeks to ensure that the encoder-then-decoder mapping $x \mapsto D_{\theta}(E_{\psi}(x))$ is as close to the identity map as possible; the sampling is done at run time from the empirical distribution $\mathbb{P}^{\text{real}}$ of objects available (e.g., for MNIST or IMAGENET this will be the empirical probability law of all images in the dataset). This gives the term: $E_{x \sim \mathbb{P}^{\text{real}}} [\|x - D_{\theta}(E_{\psi}(x))\|^2]$. a variational part that ensures that, when the empirical distribution $\mathbb{P}^{\text{real}}$ is passed through the encoder E_{ϕ} , we recover the target distribution, denoted here $\mu(dz)$ that is usually taken to be a Multivariate normal distribution. We will denote $E_{\phi} \# \mathbb{P}^{\text{real}}$ this pushforward measure which in practice is just the empirical distribution obtained by passing all dataset objects through the encoder E_{ϕ} . In order to make sure that $E_{\phi} \# \mathbb{P}^{\text{real}}$ is close to the target $\mu(dz)$, a Statistical distance d is invoked and the term $d(\mu(dz), E_{\phi} \# \mathbb{P}^{\text{real}})^2$ is added to the loss. We obtain the final formula for the loss: